

Name..... Adm. No..... Sec.....

1. Choose the correct statement/statements: There exists a binomial distribution with

- A) mean=3 and variance=4 ☒ B) mean=2 and variance=1 C) mean=3 and variance=3 D) mean=1 and variance=2

mark: 1

2. For a normal distribution, the first moment about 10 is 40 and the fourth moment about 50 is 48. What is the mean and standard deviation of the distribution?

- A) 30,3 B) 40,2 ☒ C) 50,2 D) 40,4

mark: 1

3. If the probability of hitting a target is 10% and 10 shots are fired independently. The probability that the target will be hit at least once is:

- A) 0.6324 B) 0.5674 (C) 0.6253 ☒ (D) 0.6513

mark: 1

4. The number of arrivals of customers during any day follows Poisson distribution with a mean of 5. The Probability that the total number of customers on two days selected at random is less than 2 is:

- ☒ (A) 0.0004994 B) 0.002543 C) 0.00768 D) 0.004564

mark: 1

5. A beta distribution is reduced to the Uniform distribution if

- A) $m=n=2$ B) $m < n$ ☒ (C) $m = 1, n = 1$ D) $m > n$

mark: 1

6. Nigel is playing tournament chess against a computer program and the probability he wins against the program at any given game is 0.25. Nigel is playing several practice games every day, one after the other. Find the probability that on a given day Nigel wins for the first time on the 4-th game played.

- A) 0.456 ☒ (B) 0.105 C) 0.011 D) 0.316

mark: 1

7. The probability of a fatal accident with a worker in an explosive industry employing 300 workers during the year is 0.005. If 1000 such industries are in existence, find the number of industries in which there will be at least two fatal accidents in the year.

- ☒ (A) approximately 442 B) approximately 960 C) approximately 450 D) approximately 300

mark: 2

8. In National Basketball Association championship series, the team that wins four games in a series of seven games, is the winner. Suppose that teams A and B face each other in the championship games and that team A has probability 0.55 of winning a game over team B. What is the probability that team A will win the series in 6th game ?

- A) 0.6083 B) 0.5931 ☒ (C) 0.1853 D) 0.1647

mark: 2